

Geometry Honors Unit 11 Assessment Guide

General Information	In unit 11, students apply the concepts learned in unit 10 to solve problems involving triangles and quadrilaterals inscribed in a circle (MA.912.GR.6.3). The content also extends to solving problems involving triangles using measures of interior angles of a triangle sum to 180°; measures of a set of exterior angles of a triangle sum to 360°; triangle inequality theorem; base angles of isosceles triangles are congruent; the segment joining midpoints of two sides of a triangle is parallel to the third side and half the length; and the medians of a triangle meet at a point. Students have used or learned several of these relationships in their prior learning or earlier in Geometry. To assess their understanding, students will apply their understanding to setting up and solving equations. The equations that students solve match MA.7.AR.2.2 or MA.8.AR.2.1. If students are struggling solving equations as found in unit 11 and on the unit assessment, teachers are encouraged to support students using grade 7 unit 10 and/or grade 8 unit 8. Question 4 shows students two different incomplete constructions. These constructions are sufficient for the student to know the location of the point of concurrency. The assessment limits for benchmark MA.912.GR.5.3 state: • Items will assess constructions through the use of compass and straight-edge with descriptions, images, and/or animations. • Items may ask the student to describe steps in a construction, identify the next step in a given construction, or describe what construction is being created. In question 4, students use their knowledge of constructions of inscribed and circumscribed circles of a triangle to complete statements. The last statement connects the construction to the measures of interior angles of a triangle sum to 180°. Question 4 helps teachers understand how constructions could be assessed when students do not have access to geometric tools to perform constructions. Question 11 assesses student understanding of MA.912.GR.5.4, construction of a regular hexagon that is inscribed in a circle.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.912.GR.6.3
$n =$ 12		Recommended Scoring (7 Total Points)	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of n .
Question	1b	Benchmark(s)	MA.912.GR.6.3
$m \widehat{PY} =$ 78°		Recommended Scoring (6 Total Points)	Consider giving students full credit if a student makes an error in 1a but correctly finds $m \widehat{PY}$.
Question	2	Benchmark(s)	MA.912.GR.1.3
Least $\angle L$ $\angle S$ Greatest $\angle T$		Recommended Scoring (6 Total Points)	No partial credit is suggested.

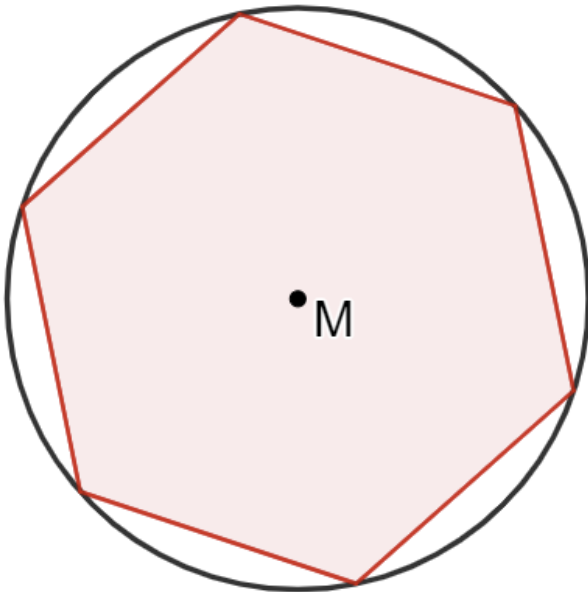
Question	3a	Benchmark(s)	MA.912.GR.1.3
RC = 35.9		Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer.
Question	3b		
Perimeter 71.3		Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer.

Question	4	Benchmark(s)	MA.912.GR.5.3
Complete the statements. ☐ Nate's ☐ Whitney's construction uses ☐ medians ☐ angle bisectors ☐ segment bisectors ☐ perpendicular bisectors to construct the incenter of $\triangle HBX$. When completed, ☐ Nate's ☐ Whitney's construction will result in a circle that is circumscribed about $\triangle HBX$ with ☐ $\overline{BN}$ ☐ $\overline{BW}$ ☐ $\overline{LN}$ ☐ $\overline{LW}$ as a radius. ☐ Nate's ☐ Whitney's construction can be used to show the Triangle Sum theorem because the angles of $\triangle HBX$ are ☐ central angles ☐ inscribed angles ☐ angles formed by tangents of the ☐ inscribed ☐ circumscribed circle such that the three arcs intercepted by the interior angles of $\triangle HBX$ complete a circle, so the sum of the measures of the interior angles of $\triangle HBX$ is half the measure of a circle.		Recommended Scoring (14 Total Points)	Consider giving students points as follows: • Correctly completes the first statement – 4 points. • Correctly completes the second statement – 4 points. • Correctly completes the third statement – 6 points. It is not suggested that students receive partial credit by further dividing up the points.

Question	5a	Benchmark(s)	MA.912.GR.6.3
$x =$ 17		Recommended Scoring (7 Total Points)	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of x .
Question	5b	Benchmark(s)	MA.912.GR.6.3
$y =$ 5		Recommended Scoring (7 Total Points)	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of y .

Question	6	Benchmark(s)	MA.912.GR.1.3
$d =$ 17.1		Recommended Scoring (7 Total Points)	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of d .
Question	7	Benchmark(s)	MA.912.GR.1.3
☐ 14, 14, 14 ☐ 17, 18, 35 ☐ $\frac{37}{4}, \frac{37}{8}, \frac{37}{12}$ ☐ 23.1, 45.6, 73.4 ☐ 45.62, 65.76, 111.21		Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correct answer. If students have any incorrect choices, then consider deducting 2 points for each incorrect answer.

Question		8a,b	Benchmark(s)	MA.912.GR.1.3						
<table><tr><th>Segment</th><th>Length</th></tr><tr><td>$\overline{RW}$</td><td>20.125</td></tr><tr><td>$\overline{RH}$</td><td>24.74</td></tr></table>			Segment	Length	$\overline{RW}$	20.125	$\overline{RH}$	24.74	Recommended Scoring (8 Total Points)	Consider giving students 4 points for each length. Consider following student work to determine if a minor error is carried through, thereby resulting in a different answer.
Segment	Length									
$\overline{RW}$	20.125									
$\overline{RH}$	24.74									
Question		9	Benchmark(s)	MA.912.GR.1.3						
☐ $\overline{YH} \cong \overline{HB}$ ☐ $\overline{YR} \cong \overline{BW}$ ☐ $\triangle RHW \sim \triangle YHB$ ☐ $\angle HRW \cong \angle HWR$ ☐ $\angle HWR \cong \angle WBY$			Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correct answer. If students have any incorrect choices, then consider deducting 2 points for each incorrect answer.						
Question		10	Benchmark(s)	MA.912.GR.6.3						
☐ $m\angle LXC = 90^\circ$ ☐ $m\angle CNH = 90^\circ$ ☐ $m\angle XNY + m\angle YLX = 180^\circ$ ☐ $m\angle MWC + m\angle MHC = 180^\circ$ ☐ $m\angle MWC + m\angle WMH = 180^\circ$			Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correct answer. If students have any incorrect choices, then consider deducting 2 points for each incorrect answer.						

Question	11	Benchmark(s)	MA.912.GR.5.4
		Recommended Scoring (6 Total Points)	No partial credit is suggested.